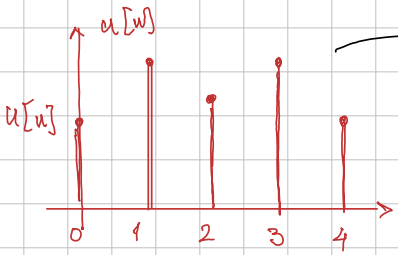
Odziv diskretnih lin. sustava

Konvolucijski zbroj

Znači:



možemo zapisati kao

$$u[0]\delta[n] + u[1]\cdot\delta[n-1] + u[2]\delta[n-2] + \dots$$

$$= \sum_{m=-\infty}^{\infty} u[m]\cdot\delta[n-m]$$

$$y[n] = \sum_{m=-\infty}^{\infty} u[m]\cdot h[n-m] = u[n]\cdot h[n] \rightarrow \text{konvolucija!}$$

Primjer $h[n] = 5n+4$

ako je na ulazu 1 impuls $\delta[n]$, na izlazu je $5n+4$

$u[n] = 2\delta[n] - 3\delta[n-1]$

$y[n] = ?$

$$y[n] = 2\cdot(5n+4) - 3\cdot(5(n-1)+4)$$

$$= 10n+8-15n+3 = -5n+11$$

Ako su i ulaz i izlaz kauzalni: ?

$u[n] = 0, \quad n<0 \qquad h[n] = 0, \quad n<0$

$$y[n] = \sum_{m=-\infty}^{\infty} u[m]\cdot h[n-m] = \underbrace{\cancel{\sum_{m=-\infty}^{-1} u[m]\,h[n-m]}}_{\substack{\text{negativni } m \\ \rightarrow u=0}} + \sum_{m=0}^{n} u[m]\cdot h[n-m] + \underbrace{\cancel{\sum_{m=n+1}^{\infty} u[m]\,h[n-m]}}_{\substack{h[n-m]<0 \\ \rightarrow h=0}}$$

$$y[n] = \sum_{m=0}^{n} u[m]\cdot h[n-m]$$

Primjer: $y[n] - 0.75\, y[n-1] = u[n]$

$$h[n] = 0.75^n \mu[n] = \begin{cases} 0.75^n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad \text{(kauzalan)}$$

$u[n] = 0.5^n \mu[n]$ (kauzalan) $\rightarrow$ budući da su. dva kauzalna koristimo $\sum_{m=0}^{n}$

$$y[n] = \sum_{m=0}^{n} u[m] \cdot h[n-m] = \sum_{m=0}^{n} 0.5^m \, 0.75^{n-m} = 0.75^n \sum_{m=0}^{n} 0.5^m \cdot \frac{1}{0.75^m} =$$

$$= \left(\frac{3}{4}\right)^n \underbrace{\sum_{m=0}^{n} \left(\frac{2}{3}\right)^m}_{\text{konačni geom. red}} = \frac{3}{4} \cdot \frac{1 - \left(\frac{2}{3}\right)^{m+1}}{1 - \frac{2}{3}} = \frac{3}{4} \cdot \frac{1 - \left(\frac{2}{3}\right)^m \cdot \left(\frac{2}{3}\right)}{\frac{1}{3}}$$

$$= \frac{9}{4}\left(1 - \left(\frac{2}{3}\right)^m \cdot \frac{2}{3}\right) = \frac{9}{4} - \frac{3}{2}\left(\frac{2}{3}\right)^m = \frac{9}{4} - \left(\frac{2}{3}\right)^{m-1}$$

Primjer: Sustav konačnog impulsnog odziva = FIR

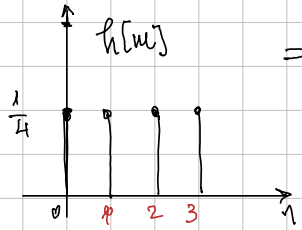
finit impulse response

$$y[n] = \frac{1}{4} \sum_{m=0}^{3} u[n-m]$$

$$= \frac{1}{4} u[n] + \frac{1}{4} u[n-1] + \frac{1}{4} u[n-2] + \frac{1}{4} [n-3]$$

$u[n] = \delta[n]$

$$h[n] = \frac{1}{4} \sum_{m=0}^{3} \delta[n-m] = \frac{1}{4}\delta[n] + \frac{1}{4}\delta[n-1] + \frac{1}{4}\delta[n-2] + \frac{1}{4}\delta[n-3]$$



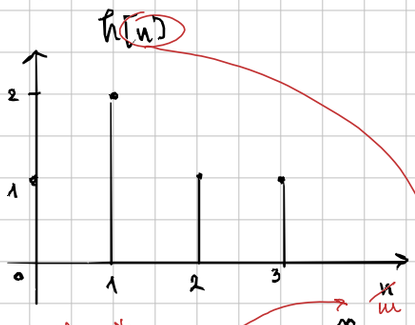
$\Rightarrow$ FIR sustav

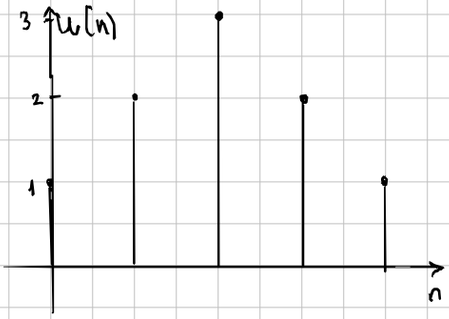
Ako znamo $h[n]$ onda nas zanima odziv za bilo koju drugu pobudu

$$y[n] = \sum_{m=0}^{M} h[m] \cdot u[n-m]$$

$$= \sum_{m=0}^{3} h[m] \cdot u[n-m] = \underset{\frac{1}{4}}{h[0]} \cdot u[n] + \underset{\frac{1}{4}}{h[1]} \cdot u[n-1] + \underset{\frac{1}{4}}{h[2]} \cdot u[n-2] + \underset{\frac{1}{4}}{h[3]} \cdot u[n-3]$$

Primjer: Kako naći odziv sustava, bilo kakvog, ali koristeći grafičku metodu → radi jednostavnosti FIR

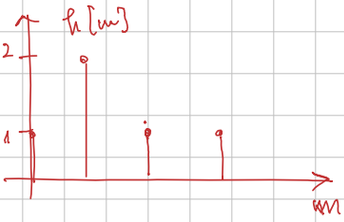




pretvoriti n u m (just like that)

$$y[n] = \sum_{m=-\infty}^{\infty} h[m] \cdot u[n-m]$$

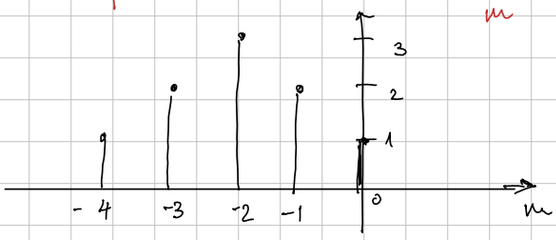
n je konstanta, trebamo u[m]



ali trebamo $\underline{\underline{-m}}$

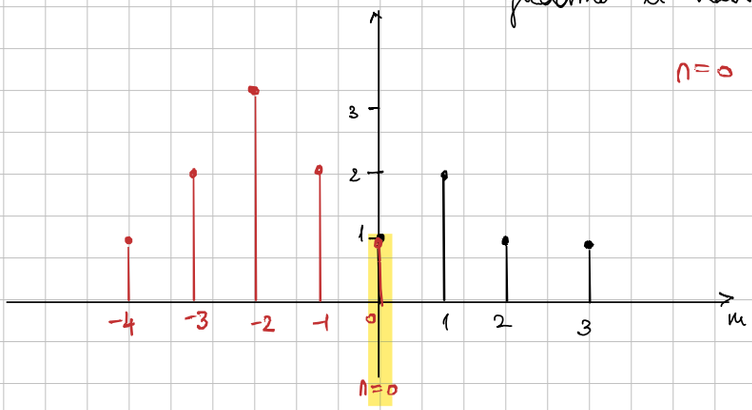
+

zrcaljenje oko y



[n−m] koliko čemo u[−m] pomicati

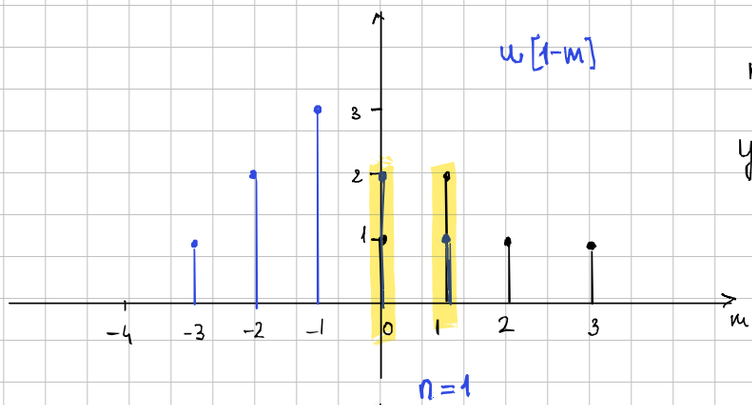
→ tražimo sve moguće pomake → crtamo mnogo slika za sve pomake i gledamo u kakvom su odnosu



n=0

(prema def konvolucije)

$y[0] = 1 \cdot 1$

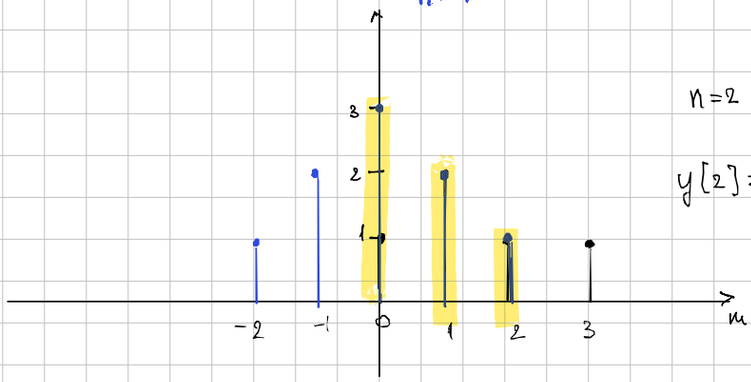


n=1

$y[1] = \cdots 3 \cdot 0 + 2 \cdot 1 + 1 \cdot 2 \ldots$

$y[1] = 4$

n=1



n=2

$y[2] = 3 \cdot 1 + 2 \cdot 2 + 1 \cdot 1$

$y[2] = 8$

→ i sad to ponavljamo dok cijeli crveni signal ne izađe van desno

ako nastavimo istim postupkom dalje:

... $y[0]=1$ $y[1]=4$ $y[2]=8$ $y[3]=11$ $y[4]=10$

$y[5]=7$ $y[6]=3$ $y[7]=1$ $\longrightarrow y[n]$

*opet smo u varijabli n

ovaj postupak je provedljiv ako su ulaz / odziv malog broja ponavljanja

SVOJSTVA konvolucije

1. **Komutativnost** $y[n]=\sum_{m=-\infty}^{\infty} h[m]\cdot u[n-m] = \sum_{m=-\infty}^{\infty} h[n-m]\cdot u[m]$

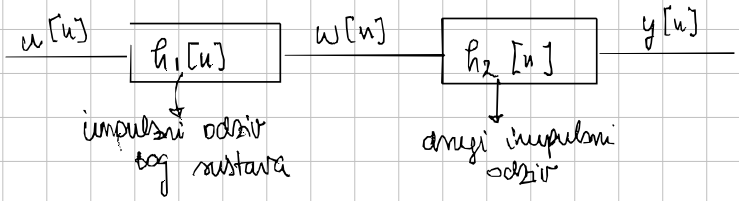
*svejedno koji zrcalimo, a kojeg ostavljamo istim

DOKAZ:

$n-m=k;\ m=n-k \Longrightarrow \sum_{k=-\infty}^{\infty} h[n-k]\cdot u[n-n+k] = \sum_{k=-\infty}^{\infty} h[n-k]\cdot u[k]$

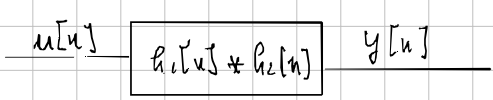
$y[n]=u[n]*h[n]=h[n]*u[n]$

2. **Asocijativnost**

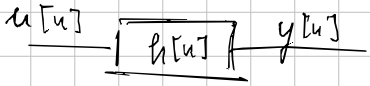


$\longrightarrow$ možemo prvo $u[n]*h_1[n]=w[n]$ i onda $w[n]*h_2[n]=y[n]$

ili



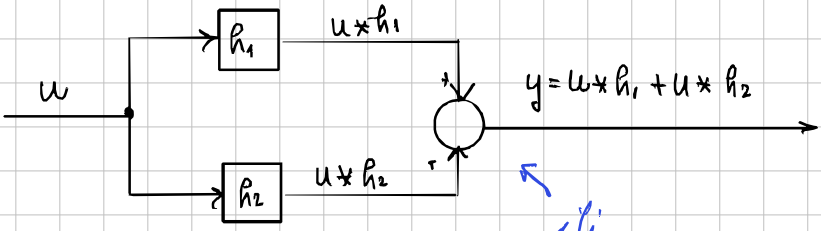
$\longrightarrow$ prvo konvoliramo dva impulsna odziva $y[n]=u[n]*\underbrace{(h_1[n]*h_2[n])}_{h[n]}$



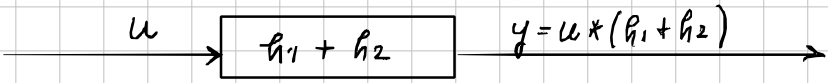
rezultat oba $y[n]$ mora biti isti da bi vrijedilo

SVOJSTVA konvolucije

3. Distributivnost = spajanje dva sustava u paralelu



ili



DOKAZ:

$$y[n] = u[n] * (h_1[n] * h_2[n]) = \sum_{m=-\infty}^{\infty} u[m]\cdot\left(h_1[n-m] + h_2[n-m]\right)$$

$$= \sum_{m=-\infty}^{\infty} u[m]\cdot h_1[n-m] + \sum_{m=-\infty}^{\infty} u[m]\cdot h_2[n-m] = u[n]*h_1[n] + u[n]*h_2[n]$$

4. Svojstvo pomaka

$$y[n] = x_1[n] * x_2[n]$$

$$y_1[n] = x_1[n-p] * x_2[n-q]$$

dokaz

$$y_1[n] = \sum_{m=-\infty}^{\infty} x_1[m-p]\cdot x_2[n-m-q] = \left| \begin{matrix} j = m-p \\ m = j+p \end{matrix} \right| = \sum_{j=-\infty}^{\infty} x_1[j]\cdot x_2[n-j-p-q]$$

* radimo samu po joj sada!

$$\Rightarrow \underbrace{\sum_{j=-\infty}^{\infty} x_1[j]\cdot x_2[\boxed{n-p-q}-j]}_{\text{definicija konvolucije}} = y[n-p-q]$$

izlaz je pomaknut za ukupni pomak s ulaza

5. Konvolucija s jediničnim impulsom

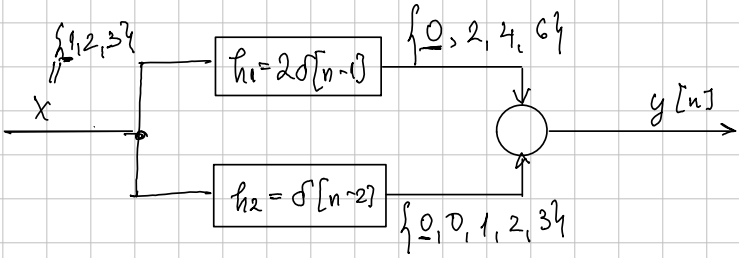
$$x[n] * \delta[n] = \sum_{m=-\infty}^{\infty} x[m]\cdot\delta[n-m] \xrightarrow[m=n]{} x[n]$$

(samo pomiče)

Primjer: Impulsni odziv 1. sustava: $h_1 = 2\delta[n-1]$

Impulsni odziv 2. sustava: $h_2 = \delta[n-2]$

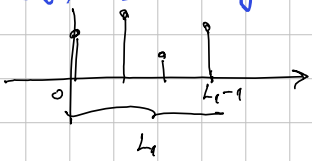
Ulazni signal (dva sustava jer paralela): $x[n] = \{1, 2, 3\}$



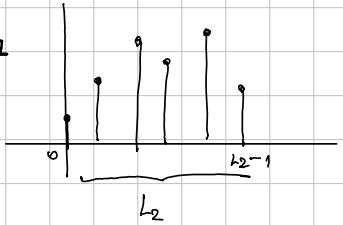
$y[n] = \{0, 2, 5, 8, 3\}$

▸ Duljina konvolucijskog zbroja

$x_1[n] \longrightarrow L_1$



$x_2[n] \longrightarrow L_2$



koliko će biti duljačak i koji je redni br. zadnjeg uzorka u y koji je nastao konvolucijom

$$y[n] = x_1[n] * x_2[n]$$

$$= \sum_{m=-\infty}^{\infty} x_1[m] \cdot x_2[n-m]$$

x_1: $0 \le m \le L_1 - 1$

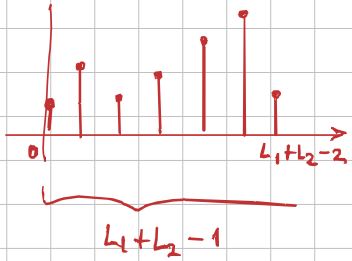
x_2: $0 \le n-m \le L_2 - 1 \quad / +m$

$m \le n \le L_2 - 1 + m$

$0 \le m \le n \le L_2 - 1 + m$

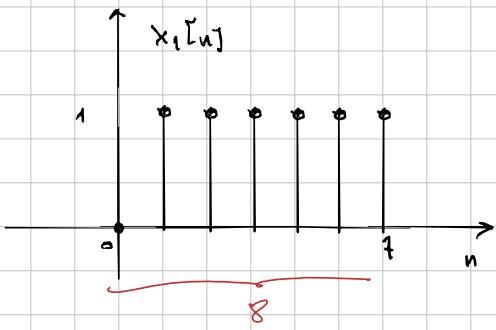
$\le L_2 - 1 + L_1 - 1$

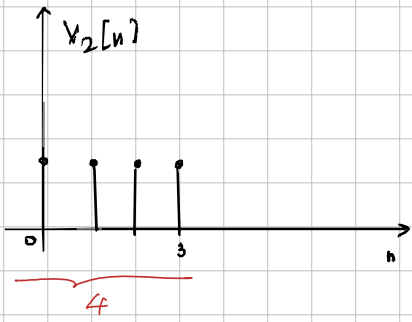
$\Rightarrow 0 \le n \le L_1 + L_2 - 2$



Primjer: konvolucija dva signala

$$x_1[n] = \begin{cases} 1, & 1 \le n \le 7 \\ 0, & \text{inače} \end{cases} \qquad x_2[n] = \begin{cases} \frac{1}{2}, & 0 \le n \le 3 \\ 0, & \text{inače} \end{cases}$$





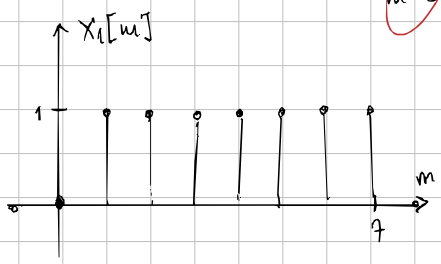
njihovo rješenje ima $8+4-1 = 11$ uzoraka

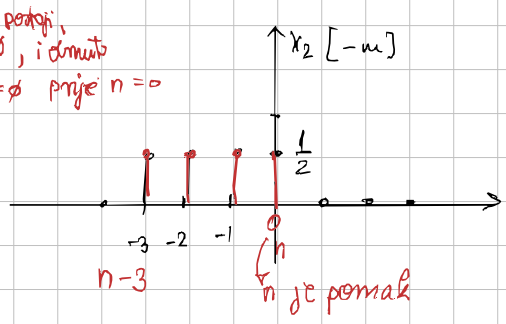
Matlab 26:00 [SIS 40.]

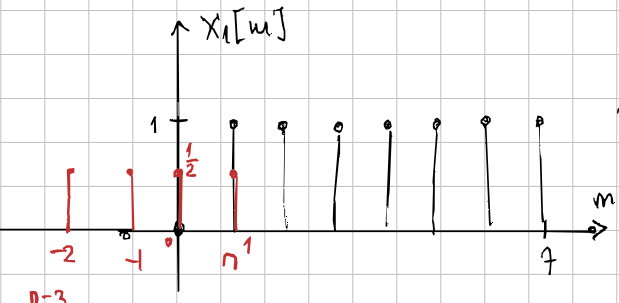
oba su kauzalni i kreću od 0 (možemo koristiti konvoluciju za kauzalne)

$$y[n] = x_1[n] * x_2[n] = \sum_{m=0}^{n} x_1[m] \cdot x_2[n-m]$$

kad crveni postoji, crni je $\emptyset$, i obrnuto
$\Rightarrow$ konv $= \emptyset$ prije $n=0$

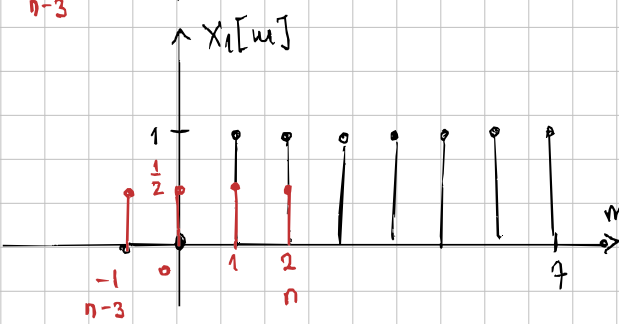






$\underline{n=1}$ prvi put

$n-3=1$ posljednje preklapanje
$\underline{n=4}$



Djelomično preklapanje
$n \in [1, 4\rangle$

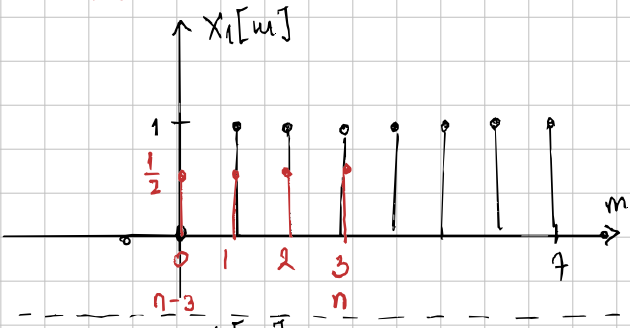
$$y[n] = \sum_{m=1}^{n} x_1[m] \cdot x_2[n-m]$$

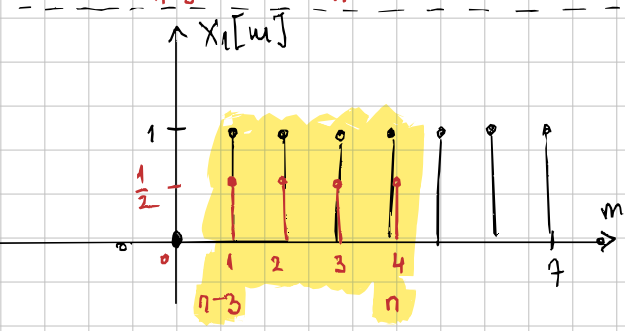
$\rightarrow$ zadnji el. parnog

prvi element mirnog

$$y[n] = \sum_{m=1}^{n} 1 \cdot \frac{1}{2} = \frac{1}{2} \cdot n$$

uvrstimo $n \in \{1, 2, 3\}$





$n=4 \rightarrow$ potpuno preklapanje

$\rightarrow$ radimo sumu sva 4 elementa

$$y[n] = 4 \cdot 1 \cdot \frac{1}{2} = \underline{\underline{2}}$$

sve vrijednosti jednake

$n \in [4, 7]$

$$y[n] = \sum_{m=n-3}^{n} 1 \cdot \frac{1}{2} = \frac{1}{2} \cdot (4) = \underline{\underline{2}}$$

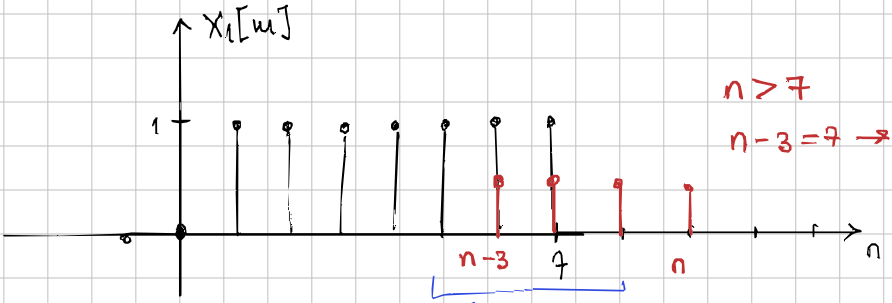
1. slučaj preklapanja:

kada je „ušao" u zajedničko:
$\hookrightarrow n-3=1 \longrightarrow n=4$

Zadnji slučaj preklapanja

$\rightarrow$ zadnji unutarnji element $\rightarrow n=7$

Nakon potpunog preklapanja crveni izlazi van:



$n > 7$

$n-3=7 \rightarrow n=10$

Zadnje preklapanje

Kad je $n=11$, cijeli crveni je izvan

Djelomično preklapanje $n \in [8,10]$.

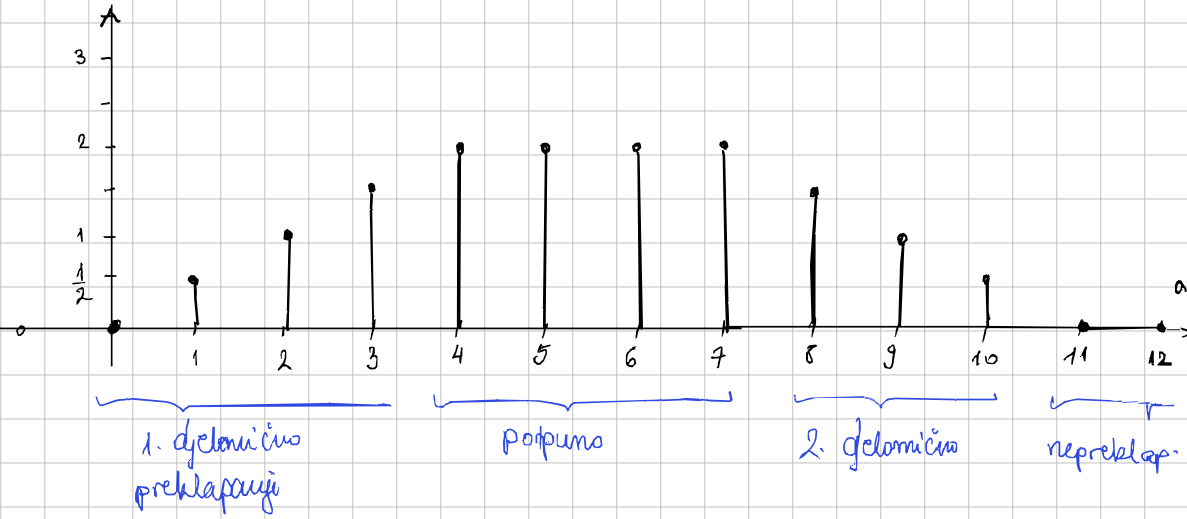
Nepreklapanje za $n \geq 11 \rightarrow y[n]=0$

$$y[n] = \sum_{m=n-3}^{7} 1 \cdot \frac{1}{2} = \frac{1}{2} \cdot \left(7-(n-3)+1\right) = \underline{\underline{\frac{1}{2}(11-n)}}$$

$7 \rightarrow$ zadnji zeleni

na tom djelu vremena izlaz tek mijenja izlaz

▸ Konačni izlaz:



$$y[n] = \begin{cases} \frac{1}{2}n, & n \in [1,4\rangle \\ 2, & n \in [4,7\rangle \\ \frac{1}{2}(11-n), & n \in [8,10] \\ 0, & \text{inače} \end{cases}$$